MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCECC102 - COMPUTER FUNDAMENTALS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Scheme of Evaluation

Part A
Very Short Answer Questions
Each Question carries one mark
Answer all questions.

1	**List different forms of input that a computer accepts.** Any two, each carrying .5 marks (Data,programs &user reply.)	[K]	[1]
2	**Write the type of computer that is used as a web server.** Any one type carries one mark (Minicomputers and above)	[K]	[1]
3	**Write the registers used for communication between the CPU and memory.** MAR & MDR (.5 marks each)	[U]	[1]
4	**Identify the bus which can transfer only one bit at a time**. Serial bus	[K]	[1]
5	**Which is the smallest addressable unit?** Byte	[U]	[2]
6	**List two write -only optical storage devices.** CD-R,CD-RW (.5 marks each)	[U]	[2]
7	**State the use of central keys in the keyboard.** Controls the movement of the cursor and screen display	[K]	[3]
8	**List the areas in which a Joystick is mostly used .** Any one carries one mark (Computer games, CAD, etc)	[K]	[3]
9	**Name the camera that store images in JPEG format.** Compact Digital Cameras	[K]	[3]
10	**Write the factors that determine the quality of video taken through webcam**. Frame rate & resolution (Each carries .5 marks)	[K]	[3]

(10 × 1 = 10)

Part B

Short Answer Questions
Answer any 4 questions.
Each question carries 5 marks

11 **Distinguish between an analog and a digital computer.** [U] [1]
One mark for each comparison

12 **Show the steps involved in memory read operation, with a diagram.** [K] [1]
Block diagram - 2 marks
Steps - 3 marks

13 **List important properties of optical storage system.** [K] [2]
Introduction - 2 marks
Properties - 3 marks

14 **Discuss the importance of output devices, and identify different types.** [U] [3]
Introduction - 2 marks
Types - 3 marks

15 **Define the characteristics of various scanning devices.** [K] [3]
Introduction - 2 marks
Characteristics - 3 marks (Any three, one mark each)

16 **Explain the different categories of computer terminals.** [U] [3]
Introduction - 2 marks
Categories - 3 marks (Any three, one mark each)

$(4 \times 5 = 20)$

Part C

Essay Questions
Answer any 2 question.
Each question carries 10 marks

17 **Classify the various types of computers based on their operating principles and describe their key characteristics.** [U] [1]
Introduction and listing the categories- 2 marks
Analog computers- 3 marks
Digital Computers-3 marks
Hybrid Computers-2 marks

18 **Explain the purpose of RAM in a computer system. Differentiate between the various types of RAM, providing examples and discussing the specific characteristics and use cases for each type.** [U] [2]
RAM introduction - 2 marks
Static RAM - 4 marks
Dynamic RAM - 4 marks

19 **Explain the various types of digital cameras, comparing their main characteristics and uses.** [U] [3]
Introduction - 2marks
Types - 6 marks
Comparison - 2 mark

$(2 \times 10 = 20)$